

Office: 601 M. L. King Jr Dr.
Computer Science Building 3205
Winston-Salem State University
Winston-Salem, NC 27110
Phone: 336-750-3323

Jinsuk Baek

E-mail: baekj@wssu.edu
URL: jinsuk-baek.github.io
Mobile: 336-402-6196

Lab: 601 M. L. King Jr Dr.
Computer Science Building 4114
Winston-Salem State University
Winston-Salem, NC 27110

Research Interests

- 1) Quantum Computing and Quantum Networking
- 2) Intelligent Networked Systems and IoT
- 3) AI-Enabled Robotics and Cyber-Physical Systems

Education

- Ph.D. Computer Science, University of Houston, Texas, USA**
August 1999 – December 2004
Dissertation: Scalable Approaches for Tree-Based Reliable Multicast
Advisor: Dr. Jehan-François Pâris
- M.Sc. Computer Science and Engineering, Hankuk University of Foreign Studies, KOREA**
March 1996 – February 1998
Thesis: An Efficient Parameter Coding Method for Object-Oriented Coder (MPEG-4)
Advisor: Dr. Sangchul Kim
- B.Sc. Computer Science and Engineering, Hankuk University of Foreign Studies, KOREA**
March 1992 – February 1996

Professional Experience

- Professor** (July 2016 – Present)
Department of Computer Science, Winston-Salem State University, Winston-Salem, NC, USA
- Associate Professor** (July 2010 – June 2016)
Department of Computer Science, Winston-Salem State University, Winston-Salem, NC, USA
- Assistant Professor** (July 2005 – June 2010)
Department of Computer Science, Winston-Salem State University, Winston-Salem, NC, USA
- Post Doctorate Research Associate** (January 2005 – June 2005)
Distributed Multimedia Research Group, University of Houston, Houston, TX, USA
Advisor: Dr. Jehan-François Pâris
- Consulting Expert** (March 2005)
Rong Gabello Law Firm for Apple, Inc.
- Research Assistant** (August 2002 – December 2004)
Distributed Multimedia Research Group, University of Houston, Houston, TX, USA
Advisor: Dr. Jehan-François Pâris
- Teaching Assistant** (August 2000 – July 2002)
Department of Computer Science, University of Houston, Houston, TX, USA
- Software Developer** (January 1999 – June 1999)
IBM Korea conjunction with Leading Soft. Co, Seoul, KOREA
Designed and implemented electronic approval systems for Intranet by using LOTUS NOTE R4.
- Research Assistant, Multimedia Information Telecommunication Lab** (March 1996 – February 1998)
Department of Computer Science, Hankuk University of Foreign Studies, KOREA
Advisor: Dr. Sangchul Kim

Awards

- Outstanding Paper Award, the 28th IEEE International Conference on Advanced Communication Technology, February 2026.
- Research Award, the HBCU Smart City Challenge, February 2023.
- Outstanding Paper Award, the 22th IEEE International Conference on Advanced Communication Technology, February 2020.
- Best Paper Award, the 2018 International Conference on Recent Advances in Engineering and Technology, October 2018.
- Outstanding Paper Award, the 18th IEEE International Conference on Advanced Communication Technology, February 2016.
- Outstanding Paper Award, the 14th IEEE International Conference on Advanced Communication Technology, February 2012.
- Professional Development Travel Grant, Winston-Salem University, 2006–2019.
- Professional Development Research Grant, Winston-Salem University, 2017–2019.
- Research Advisor for 2010-2011 NC Space Grant Graduate Research Fellowship
- Research Advisor for 2007-2008 NC Space Grant Graduate Research Fellowship
- Research Advisor for 2006-2007 NC Space Grant Graduate Research Fellowship
- Encouragement Prize, the 1st SAMSUNG Electronics Future Technology Research Contest, 2000.
- Infrastructure Studies Student in Japan, Sponsored by SAMSUNG, 1994.

Teaching Experience

1) Courses Developed and Taught

Undergraduate Level

- Directed Study (CSC 4390, Spring 2025, Fall 2026)
- Computer Communication Networks (CSC 4389, Spring 2006–2026)
- Hardware and Media Security (CSC 4360, Fall 2023)
- Cryptography (CSC 4330, Fall 2007–2008, Spring 2016–2019)
- Parallel Computing (CSC 4322, Spring 2010, 2012, Fall 2025)
- Analysis of Algorithms (CSC 3331, Spring 2025–2026)
- Computer Architecture (CSC 3322, Spring 2007, 2009, 2019–2026)
- Operating Systems (CSC 3321, Fall 2009–2011, Spring 2026)
- Intr. to Computer Hardware Organization (CSC 2320, Fall 2012, 2014–2022, Spring 2020–2022)
- Intr. to Computer Software Systems (CSC 2310, Spring 2011, 2013–2016)
- Professional Development Seminar (CSC 2131, Fall 2024)
- Intr. to Computer Technology (CSC 1307, Fall 2013)
- Computer and Its Use (CSC 1306, Fall 2005–2007, Fall 2010–2012, 2014, Summer 2006–2009)

Graduate Level

- Network Security (CST 6309, Spring 2006, Spring 2010–2015)
- Network Design, Performance and Evaluation (CST 6308, Spring 2008–2009)
- Advanced Computer Architecture (CST 6301, Fall 2005–2009, Spring 2006, Fall 2013–2026)
- Cryptography (CST 5335, Fall 2007–2008, Spring 2016–2019)
- Computer Networks (CST 5328, Fall 2005–2026)
- Parallel Computing (CST 5301, Spring 2010, 2012, Fall 2024–2025)

Advising

1) Master's Thesis/Project Advisor

- Shymik Smith, "Robust Pick-and-Place Pipeline for Collaborative Robotic Arms: Toward Autonomous Unmanned Kitchen Systems," May **2026. (Project)**
- Kevin A. Rodriguez Fuentes, "ShuttleFlow: Improving Shuttle Tracking Around Campus," December **2025. (Project Co-advisor)**
- Emmanuel White, "Development of an E-Commerce Platform," May **2025 (Project Co-advisor)**
- Pamela Baucum, "Development of Framework to Control Bluetooth-Based Robot Car Using an Arduino Microcontroller," May **2020. (Project)**
- Jazmyn Saunders, "Design and Development of a Remote Door Locking System Using Arduino," May **2019. (Project)**
- Hitesh Pambhar, "An Implementation for Framework for Remote Alarm System Using an Arduino Microcontroller," November **2018. (Project)**
- Magdeldin Abdelrahman, "Performance Evaluation of Known Class of TCP using ns-2 Simulation," WSSU, November **2014. (Project)**
- Bendhi Eliassaint, "Performance Evaluation of TCP with Various Traffic Patterns using ns-2," WSSU, May **2014. (Project)**
- Kristyn Daney, "Implementation of Round Trip Time Simulator with ns-2," WSSU, May **2012. (Project)**
- Howard Brown, "A Backoff Scheme for Multicasting in IEEE.802e Wireless Networks," WSSU, May **2012. (Thesis)**
- Olasumbo Adeyeye, "Numerical Performance Analysis for Data Aggregation in Wireless Sensor Networks" WSSU, May **2011. (Project)**
- Edward Jordan, "Operations of BS with Mobile Robot for Wireless Sensor Networks," WSSU, December **2010 (Thesis)**
- Basheer Olusegun Idowu, "Implementation of (N, 1) Steganography Approach for Secret Sharing," WSSU, April **2010. (Project)**
- Tarandeep Kaur, "Energy Efficient Deployment and Cluster Formation for Wireless Sensor Networks," WSSU, November **2008. (Thesis)**
- Geethanjali Ramaiah, "On Demand Multimedia Service using an Advanced Multicast Technique," WSSU, November **2008. (Project)**
- Derrick Hargrove, "An Urgent NAK with Prediction of Lost Packets for Fast Error Recovery in a Reliable Multicast," WSSU, May **2008. (Thesis)**
- Kaur Gurpreet, "A New Hybrid Caching Architecture for Cooperative Caching," WSSU, May **2007. (Thesis)**
- Wood Kanampiu, "A NAK Suppression Scheme for Group Communications Considering the Spatial Locality of Packet Losses", WSSU, December **2006. (Thesis)**

2) Master's Thesis/Project and Ph.D. Dissertation Committee Member

- Yuanjie Li, "Quantum Principles in Concurrency and Quantum Cross Hybrid Fine-Tuning Strategies for LLMs in Multi-Tier IoT Resource Allocation," February **2026. (Korea University, Dissertation)**
- Han Ying, "GRPO Reinforcement Learning for Reducing Quantum Circuit Gate Count in the NISQ Era," May **2026. (Korea University, Thesis)**
- Zarin Tasnim Shejuti, "Integrated Postural Assessment and Indoor Localization for Heathcar Workers Using Edge Devices," December **2025. (Thesis)**

- Kemai Parker, “Exploring Factors Contributing to Low Academic Performance in Winston-Salem Forsyth County Public Schools (WSFCS),” December **2023**. (**Project**)
- Anthony Smith, “Effectiveness of a Bluetooth RSSI-based Vectorized Algorithm for Mobility Detection within Indoor Space,” April **2023**. (**Thesis**)
- Jamal Pete, “Data Mining: A Comparison between Rules from Associative Learning and Finite Inductive Sequences,” November **2012**. (**Thesis**)
- Salem Alghamdi, “Secure Payment Transaction Using Two-Factor Authentication,” Winston-Salem State University, May **2011**. (**Thesis**)
- Sultan Almalki, “Credit Card Online Security,” Winston-Salem State University, May **2011**. (**Thesis**)
- Elihah Myers, “FI Algorithm: Achieving Syntactic Pattern Recognition in Finite Inductive Strings,” WSSU, April **2010**. (**Thesis**)
- Tina R. Westphal, “Performance Analysis of Wavelet Compression vs. Discrete Cosine Compression in A Medical Imaging Archive Environment,” WSSU, April **2010**. (**Project**)
- Shaun Damon Southerland, “The Benefits of Using Pay Per Click Advertising for E-Commerce,” WSSU, November **2008**. (**Project**)
- Lamonica Stone, “Data Compression,” WSSU, November **2008**. (**Project**)
- Srinivasula Reddy Iragamreddy, “Intrusion Detection Using Apriori Algorithm,” December **2007**. (**Thesis**)
- John McDonald, “Topic Maps for Mobile Computing,” WSSU, December **2006**. (**Thesis**)
- Dandan Wang, “Searching and Querying Topic Maps in TM4L,” WSSU, July **2006**. (**Thesis**)

Research Publications

Refereed Journals

- [1] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, “A Lightweight Closed-Form RGB-D Image Processing Pipeline for Real-Time Yaw Estimation in Resource-Constrained Robotic Systems,” *Journal of Real-Time Image Processing* (JRTIP) (Ready to submit).
- [2] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, “Beyond Protocol Interoperability: Task-Aware CoAP-MQTT Cooperation for Robotic Smart Kitchens,” *IEEE Network* (Submitted, Under Review).
- [3] Munene W. Kanampiu, and **Jinsuk Baek**, “Secure and Resilient Network Architecture for Smart Campus Internet of Things,” *International Journal of Informatics and Communication Technology* (Submitted, Under review).
- [4] **Jinsuk Baek**, and Minh Jo, “Can Quantum Computing Power Intelligent Systems?” *IEEE IT Professional*, (Submitted, Under review).
- [5] YuanJie Li, **Jinsuk Baek**, Shinhoo Kang, and Minh Jo, “PQA: Pure-Quantum Circuit Adaptation for Parameter-Efficient LLM Fine-Tuning,” *IEEE Transactions on Neural Networks and Learning Systems*, (Submitted, Under review).
- [6] **Jinsuk Baek**, “Workload-Aware Error Mitigation for Consumer-Oriented Quantum Cloud Computing: Practical Evaluation and Insights on NISQ Devices,” *IEEE Transactions on Consumer Electronics*, (Submitted, Under review).
- [7] YuanJie Li, **Jinsuk Baek**, Shinhoo Kang, and Minh Jo, “Control-Plane-Driven Adaptive Routing for Time-Varying Quantum Repeater Networks,” *IEEE Transactions on Network Science and Engineering* (Submitted, Under review).
- [8] **Jinsuk Baek**, and Minh Jo, “No Cloning, No Retransmission: Reliable Multicast Architecture and Open Challenges for the Quantum Internet,” *IEEE Network* (Submitted, Under review).

- [9] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, “Uncertainty-Aware RGB-D Vision Sensing for Reliable Robotic Sorting,” *IEEE Sensors Journal* (Revision Submitted, Under review).
- [10] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, “Dual-Arm Synchronization Framework for Smart Kitchen Robot Systems,” *IEEE Access*, vol. 14, pp. 134316-134331, September **2026**.
- [11] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, “RL-ACK: A Reinforcement Learning-Based Adaptive ACK Framework for Heterogeneous Smart Kitchen IoT Networks,” *IEEE Access*, vol. 14, pp. 100508–100523, July **2026**.
- [12] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, “Collaborative Multi-Robot Architecture for Smart Kitchens,” *IEEE Consumer Electronics Magazine*, pp. 1–8, April **2026**.
- [13] YuanJie Li, Kyungmin Lim, **Jinsuk Baek**, and Minh Jo, “QCHFT: Quantum Cross-Hybrid Fine-Tuning for LLMs,” *IEEE Transactions on Quantum Engineering*, vol. 7, Article no. 2101021, pp. 1–21, May **2026**.
- [14] YuanJie Li, **Jinsuk Baek**, and Minh Jo, “Quantum Principles in Concurrency: Toward Resolving Classical Synchronization Problems,” *KSII Transactions on Internet and Information Systems*, vol. 20, no. 5, pp. 2773–2796, May **2026**.
- [15] Shiva Dahal, John T. Yi, and **Jinsuk Baek**, “Hardware-Aware Quantum Education: Six Modules Bridging Theory, Hardware, and Software for Next-Generation Quantum Workforce,” *KSII Transactions on Internet and Information Systems*, vol. 20, no. 5, pp. 2661–2686, May **2026**.
- [16] Kemai Parker, Munene W. Kanampiu, and **Jinsuk Baek**, “Exploring Factors Contributing to Low Academic Performance in Winston-Salem Forsyth County Public Schools (WS/FCS),” *Information Systems*, vol. 25, no. 1, January **2024**.
- [17] Cheonshik Kim, Ching-Nung Yang, **Jinsuk Baek**, and Lu Leng, “Survey on Data Hiding Based on Block Truncation Coding,” *Applied Sciences*, Volume 11, Issue 19, 9209, October **2021**.
- [18] **Jinsuk Baek**, Munene W. Kanampiu, and Cheonshik Kim, “A Secure Internet of Things Smart Home Network: Design and Configuration,” *Applied Sciences*, Volume 11, Issue 14, 6280, July **2021**.
- [19] Paul S. Fisher, Jimmy James II, **Jinsuk Baek**, and Cheonshik Kim, “Mining Intelligent Solution to Compensate Missing Data Context of Medical IoT Devices,” *Personal and Ubiquitous Computing*, Volume 22, Issue. 1, pp. 219–224, February **2018**.
- [20] Doowon Kim, **Jinsuk Baek**, Paul S. Fisher, and Sangchul Kim, “fFTP: A Fast File Transfer Protocol for Home N-Screen Platform,” *Personal and Ubiquitous Computing*, Volume 22, Issue. 1, pp. 143–152, February **2018**.
- [21] Paul S. Fisher, Nawaf Aljohani, and **Jinsuk Baek**, “Generation of Finite Inductive, Pseudo Random, Binary Sequences,” *Journal of Information Processing Systems*, Volume 13, Issue. 6, pp. 1554–1574, December **2017**.
- [22] **Jinsuk Baek**, Cheonshik Kim, “An Energy-Efficient Video Transport Protocol for Personal Cloud-Based Computing,” *Journal of Real-Time Image Processing (JRTIP)*, Volume 12, Issue 2, pp. 303–310, June **2016**.
- [23] **Jinsuk Baek**, Min Gyung Kwak, and Paul S. Fisher, “A Smart Backoff Scheme for Multicast Flows in Wireless LAN,” *The Smart Computing Review (Smart CR)*, Volume 5, Issue 3, pp. 159–167, June **2015**. (*Invited Paper*)
- [24] Paul S. Fisher, **Jinsuk Baek**, and Howard Fisher, “APPMPS: An Asynchronous, Parallel, Plastic, Multi-Processing System,” *IEEE Systems Journal (IEEE SJ)*, Volume 9, Issue 2, pp. 382–392, June **2015**.
- [25] Yong woon Ahn, Albert M. K. Cheng, **Jinsuk Baek**, Minh Jo, and Hsiao-Hwa Chen, “An Auto-Scaling Mechanism for Virtual Resources to Support Mobile, Pervasive, Real-time, Healthcare

- Applications in Cloud Computing,” *IEEE Network* (Network), Volume 27, Issue 5, pp. 62–68, October **2013**.
- [26] Mohammad M. Fuad, Debzani Deb, and **Jinsuk Baek**, “Static Analysis, Code Transformation and Runtime Profiling for Self-healing,” *Journal of Computers* (JCP), Volume 8, Issue 5, pp. 1127–1135, May **2013**.
 - [27] **Jinsuk Baek**, Min Gyung Kwak, and Paul S. Fisher, “A Hybrid Upstream Bandwidth Allocation Method for Multimedia Communications in EPONs,” *IEEE Transactions on Smart Processing and Computing* (IEEEK SPC), Volume 1, Issue 1, pp. 27–33, August **2012**. (*Invited Paper*)
 - [28] **Jinsuk Baek**, Paul S. Fisher, Minh Jo, and Hsiao-Hwa Chen, “A Reliable Overlay Video Transport Protocol for Multicast Agents in Wireless Mesh Networks,” *Wiley International Journal of Communication Systems*, Volume 25, Issue 5, pp. 553–570, May **2012**.
 - [29] **Jinsuk Baek**, Doowon Kim, Paul S. Fisher, and Minh Jo, “An Adaptive Primary Path Switching Scheme for Seamless mSCTP Handover,” *The Smart Computing Review* (Smart CR), Volume 1, Issue 2, pp. 150–160, December **2011**. (*Invited Paper*)
 - [30] Yong woon Ahn, **Jinsuk Baek**, Albert M. K. Cheng, Paul S. Fisher, and Minh Jo, “A Fair Transmission Opportunity by Detecting and Punishing the Malicious Wireless Stations in IEEE 802.11e EDCA Network,” *IEEE Systems Journal* (IEEE SJ), Volume 5, Issue 4, pp. 486–494, December **2011**.
 - [31] **Jinsuk Baek**, Jan Fischer, Hsiao-Hwa Chen, and Minh Jo, “On a Moving Direction Pattern Based MAP Selection Model for HMIPv6 Networks,” *Elsevier Journal of Computer Communications* (Elsevier ComCom), Volume 34, Issue 2, pp. 150–158, February **2011**.
 - [32] **Jinsuk Baek**, Sun Kyong An, and Paul S. Fisher, “Dynamic Cluster Header Selection and Conditional Re-clustering for Energy Efficient Wireless Sensor Networks,” *IEEE Transactions on Consumer Electronics* (IEEE TCE), Volume 56, Issue 4, pp. 2249–2257, November **2010**.
 - [33] **Jinsuk Baek**, Paul S. Fisher, Minh Jo, and Hsiao-Hwa Chen, “A Lightweight SCTP for Partial Reliable Overlay Video Multicast Service for Mobile Terminals,” *IEEE Transactions on Multimedia* (IEEE TM). Volume 12, No. 7, Volume 12, Issue 7, pp. 754–766, November **2010**.
 - [34] Tarandeep Kaur, and **Jinsuk Baek**, “A Strategic Deployment and Cluster-Header Selection for Wireless Sensor Networks,” *IEEE Transactions on Consumer Electronics* (IEEE TCE), Volume 55, Issue 4, pp. 1890–1897, November **2009**.
 - [35] Young woon Ahn, Albert Cheng, **Jinsuk Baek**, and Paul S. Fisher, “A Multiplayer, Real Time, Game Protocol Architecture to Reduce Network Latency,” *IEEE Transactions on Consumer Electronics* (IEEE TCE), Volume 55, Issue 4, pp. 1883–1889, November **2009**.
 - [36] **Jinsuk Baek**, Kaur Gurpreet, and Junghoon Yang, “A New Hybrid Caching Architecture for Cooperative Caching,” *Journal of Ubiquitous Convergence Technology*, Volume 2, No 1, pp. 1–11, August **2008**.
 - [37] **Jinsuk Baek** and Jehan-François Pâris, “A Scalable Recovery Tree Construction Considering Packet Losses Correlation for Reliable Multicast,” *KSII Transactions on Internet and Information Systems*, Volume 2, No 2, pp. 82–102, April **2008**.
 - [38] **Jinsuk Baek**, Paul S. Fisher, and Mingyung Kwak, “FI-Based Local Group Key Generation/Distribution for Mobile Multicast in a Hierarchical Mobile IPv6 Network,” *KSII Transactions on Internet and Information Systems*, Volume 2, No 1, pp. 5–22, February **2008**.
 - [39] **Jinsuk Baek**, Cheonshik Kim, and YouSik Hong, “Packet Loss Patterns Adaptive Feedback Scheduling for Reliable Multicast,” *Journal of Ubiquitous Convergence Technology*, Volume 1, No 1, pp. 28–34, November **2007**.

- [40] **Jinsuk Baek** and Wood Kanampiu, “A NAK Suppression Scheme for Group Communications Considering the Spatial Locality of Packet Losses,” *International Journal of Computer Science and Network Security*, Volume 6, No 10, pp. 158–167, October **2006**.
- [41] **Jinsuk Baek** and Eunjung Lee, “A Dynamic Authentication, Authorization, and Accounting (AAA) Resolution for Hierarchical Mobile IPv6 Network,” *International Journal of Computer Science and Network Security*, Volume 6, No 7B, pp. 170–179, July **2006**.
- [42] **Jinsuk Baek** and Eunjung Lee, “A Dual Mode Buffer for Reliable Multicast in Mobile IP Networks,” *International Journal of Computer Science and Network Security*, Volume 6, No 5B, pp. 1–8, May **2006**.
- [43] **Jinsuk Baek** and Jehan-François Pâris, “A Heuristic Buffer Management and Retransmission Control Scheme for Tree-Based Reliable Multicast,” *ETRI Journal*, Volume 27, No 1, pp.1–12, February **2005**.

Refereed Conferences with Proceedings

- [1] **Jinsuk Baek**, and Munene W. Kanampiu, “Criticality-Aware Hybrid Networking for Scalable and Resilient Robotic Smart Kitchens,” *Proceedings of 2027 IEEE International Conference on Advanced Communication Technology (ICACT 2027)*. (under review)
- [2] **Jinsuk Baek**, “Reliable Short Control Messaging via Multi-Route Quantum Superdense Coding for Future Consumer Networks,” *Proceedings of IEEE Consumer Communications and Networking Conference (IEEE CCNC 2027)*, January 2027. (under review)
- [3] Nisha Somai, Dahyeong Jeong, **Jinsuk Baek**, and Minho Jo, “Hyperbolic Large Concept Models: Geometry-Aware Hierarchical Reasoning Beyond Euclidean Space,” Auxiliary-Branch Complexity in Mamba for Efficient LLMs,” *Proceedings of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*, December **2026**. (under review)
- [4] Kyungmin Lim, **Jinsuk Baek**, Munene W. Kanampiu, and Minho Jo, “Auxiliary-Branch Complexity in Mamba for Efficient LLMs,” *Proceedings of the 21th International Conference on Intelligent Computing (INTCOM 2026)*, July **2026**.
- [5] Kyungmin Lim, **Jinsuk Baek**, and Minho Jo, “A Teleportation-Based Quantum Repeater with Multi-Route Redundancy and Stabilizer Verification,” *Proceedings of 2026 International Conference on Artificial Intelligence, Robotics, and Control (AIRC 2026)*, April **2026**.
- [6] Munene W. Kanampiu, and **Jinsuk Baek**, “A Cooperative Retransmission Control Scheme for CoAP in Constrained IoT Networks,” *Proceedings of 2026 IEEE International Conference on Advanced Communication Technology (ICACT 2026)*, February **2026**. (**Outstanding Paper Award**)
- [7] Sunghwan Kim, I Gede Pasek Suta Wijaya, **Jinsuk Baek**, and Minho Jo, “Bayesian Deep Learning Approach for Traffic Accident Severity Prediction with Uncertainty Estimation,” *Proceedings of 2025 International Conference on Internet (ICONI 2025)*, December **2025**.
- [8] Shymik Smith, Munene W. Kanampiu, and **Jinsuk Baek**, “A Robust Pick-and-Place Pipeline for Collaborative Robotic Arms: Toward Autonomous Unmanned Kitchen Systems,” *Proceedings of 2025 International Conference on Computational Science & Computational Intelligence (CSCI 2025)*, December **2025**.
- [9] Youngwon Yu, Kyungmin Lim, Taras Maksmyuk, **Jinsuk Baek**, Vladimir Simic, and Minho Jo, “RAG-Based Generative Large Language Model for University Regulations,” *Proceedings of the 20th Asia Pacific International Conference on Information Science and Technology (APIC-IST 2025)*, July **2025**.

- [10] Hitesh Pambhar, and **Jinsuk Baek**, “Enhancing MQTT for Targeted Communication in Automated Smart Homes,” *Proceedings of 2024 International Conference on Computational Science & Computational Intelligence (CSCI 2024)*, December **2024**.
- [11] Ahmad Hijaz, Ibrahim Yakubu, and **Jinsuk Baek**, “Integrating Secure Quantum Digital Signature into Quantum Communications,” *Proceedings of 2024 International Conference on Emergent Quantum Technologies (ICEQT 2024)*, July **2024**.
- [12] Makayla Morgan, Ahmad Hijaz, and **Jinsuk Baek**, “WSSU Quantum Mini Tennis: Learning Quantum Gates and State Vectors Through Gaming,” *Proceedings of the 2024 Association of Computer Science Departments at Minority Institutions Symposium (ADMI 2024)*, April **2024**.
- [13] Teo Kim, **Jinsuk Baek**, and John Yi, “Quantum Authentication Protocol for Secure Quantum Superdense Coding,” *Proceedings of 2024 IEEE Southeast Conference (IEEE SoutheastCon 2024)*, March **2024**.
- [14] John Yi, **Jinsuk Baek**, John Merle, Xiuping Tao, and Kasra Sardashiti, “Preparing a Diverse Quantum Workforce in Carolinas through Hands-on Education and Training Experiences,” *American Physical Society March Meeting (APS 2024)*, March **2024**.
- [15] Jaylin Butts, Emmanuel White, **Jinsuk Baek**, “A Quantum Teleportation Protocol Secured by a Blockchain Technology,” *Proceedings of 2023 International Conference on Emergent Quantum Technologies (ICEQT 2023)*, pp. 1665–1669, July **2023**.
- [16] **Jinsuk Baek**, and Munene W. Kanampiu, “A MQTT-based Smart Home Monitoring System: Design and Implementation,” *Proceedings of 2022 International Conference on Wireless Sensor Networks, Ubiquitous Computing and Applications*, December **2022**.
- [17] **Jinsuk Baek**, and Munene W. Kanampiu, “A MQTT-based Soil Moisture Level Notification System for a Smart Farm,” *Proceedings of 2021 IEEE International Conference on Power and Energy Applications (ICPEA 2021)*, October **2021**.
- [18] **Jinsuk Baek**, and Munene W. Kanampiu, “A Strategic Sensor Placement for a Smart Farm Water Sprinkler System: A Computational Model,” *Proceedings of 2021 IEEE International Conference on Advanced Communication Technology (ICACT 2021)*, February **2021**.
- [19] **Jinsuk Baek**, Munene W. Kanampiu, and Byeong Hyun Ko, “A Set Notation Configuration CoAP Scheme for Internet of Things,” *Proceedings of 2020 IEEE International Conference on Advanced Communication Technology (ICACT 2020)*, February **2020**. (*Outstanding Paper Award*)
- [20] **Jinsuk Baek**, Munene W. Kanampiu, “An Evaluation of How Much HTTP/1.1 Outperforms HTTP/1.0,” *Proceedings of 2019 International Conference on Science, Technology, Engineering and Management (ICSTEM 2019)*, pp. 1–3, March **2019**.
- [21] Hitesh Pambhar, and **Jinsuk Baek**, “An Implementation for Framework for Remote Alarm System Using an Arduino Microcontroller,” *Proceedings of 2018 International Conference on Recent Advances in Engineering and Technology (ICRAET 2018)*, pp. 1–3, October **2018**.
- [22] Geethanjali Ramaiah, **Jinsuk Baek**, “Client-Based, On-Demand Multimedia Service Using an Advanced Multicast Transmission Technique,” *Proceedings of Seoul International Conference on Engineering and Applied Science (SICEAS)*, pp. 123–125, December **2017**.
- [23] **Jinsuk Baek**, Paul S. Fisher, “Packet Damage-Resistant Analysis Using a Data Mining Mechanism in Wireless Sensor Networks,” *Proceedings of 13th IEEE International Conference on Intelligent Environments (IE 2017)*, pp. 176–179, August **2017**.
- [24] **Jinsuk Baek**, Paul S. Fisher, “A Simple Broadcast-Based Service Discovery for Home-Based Lighting Control Network,” *Proceedings of Seoul International Conference on Engineering and Applied Science (SICEAS)*, pp. 455–457, February **2017**.

- [25] Jamal Pete, Paul S. Fisher, and **Jinsuk Baek**, “Reliable Decision Making: A Comparison between Rules from Associative Learning and Finite Inductive Sequences,” *Proceedings of 7th Asia-Pacific International Symposium on Advanced Reliability and Maintenance Modeling (APARM 2016)*, pp. 461–468, August **2016**.
- [26] Min Gyung Kwak, **Jinsuk Baek**, and Paul S. Fisher, “Retransmission Control Scheme for RDM Protocol for Lightning Control Networks,” *Proceedings of 2016 IEEE International Conference on Advanced Communication Technology (ICACT 2016)*, pp. 86–90, February **2016**. (**Outstanding Paper Award**)
- [27] Min Gyung Kwak, **Jinsuk Baek**, and Paul S. Fisher, “A Degree Constrained Spanning Tree-Based Device Discovery Scheme for Lightning Control Networks,” *Proceedings of 2015 IEEE International Conference on Internet of Things (IoT 2015)*, pp. 1–2, October **2015**.
- [28] Cheonshik Kim, **Jinsuk Baek**, and Paul S. Fisher, “Lossless Data Hiding for Binary Document Images Using n -Pairs Pattern,” *International Conference on Information Security and Cryptology (ICISC 2014)*, pp. 317–327, December **2014**.
- [29] Paul S. Fisher, Min Gyung Kwak, Eunjung Lee, and **Jinsuk Baek**, “A Signature Scheme for Digital Imagery,” *2014 International Conference on Information Science and Applications (ICISA 2014)*, pp. 115–118, May **2014**.
- [30] Doowon Kim, **Jinsuk Baek**, and Paul S. Fisher, “Adaptive Video Streaming over HTTP,” *2014 ACMSE conference (ACMSE 2014)*, March **2014**. (poster session)
- [31] Doowon Kim, Chaitanya Achan, **Jinsuk Baek**, and Paul S. Fisher, “Implementation of Framework to Identify Potential Phishing Websites,” *2013 IEEE Intelligence and Security Informatics (IEEE ISI 2013)*, pp. 268, June **2013**. (poster session)
- [32] Howard Brown, Min Gyung Kwak, **Jinsuk Baek**, and Paul S. Fisher, “Impact of Multicast Flow for Performance of IEEE 802.11e in Wireless LAN,” *Proceedings of 2013 IEEE International Conference on Advanced Communication Technology (ICACT 2013)*, pp. 768–773, January **2013**.
- [33] Hyunju Jo, Sangchul Kim, **Jinsuk Baek**, and Paul S. Fisher, “Motion Recognition with Smart Phone Embedded 3-Axis Accelerometer Sensor,” *IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC 2012)*, pp. 919–924, October **2012**.
- [34] Wood Kanampiu, Justin Zhan, and **Jinsuk Baek**, “A Secure Group Collaboration Protocol for Non-Verbal Human Social Signals Featuring Deception Detection,” *Proceedings of 2012 IEEE International Conference on Information Privacy, Security, Risk, and Trust (IEEE PASSAT 2012)*, pp.813–818, September **2012**.
- [35] Illyoung Choi, **Jinsuk Baek**, Sangchul Kim, and Paul S. Fisher, “A New File Systems Specialized for N-Screen Platforms,” *Proceedings of 2012 IEEE International Conference on Advanced Communication Technology (ICACT 2012)*, pp. 231–234, February **2012**. (**Outstanding Paper Award**)
- [36] Mohammad M. Fuad, Debzani Deb, and **Jinsuk Baek**, “Self-Healing by means of Runtime Execution Profiling,” *Proceedings of 2011 International Conference on Computer and Information Technology (ICCIT 2011)*, pp. 202–207, December **2011**.
- [37] Yong woon Ahn, Chaitanya Belwal, Albert M. K. Cheng, and **Jinsuk Baek**, “Improving QoS for ECG Data Transmission with Enhanced Admission Control in EDCA-Based WLANs,” *Proceedings of IEEE Global Telecommunications Conference (IEEE GLOBECOM 2011)*, pp. 1–5, December **2011**.
- [38] Edward Jordan, **Jinsuk Baek**, and Wood Kanampiu, “Impact of Mobile Sink for Wireless Sensor Network,” *Proceedings of the 49th ACM Southeast Conference (ACM SE 2011)*, pp. 328–329, March **2011**. (poster session)

- [39] Elijah Myers, Paul S. Fisher, Keith Irwin, **Jinsuk Baek**, and Joao Setubal, “The Complexity and Application of Syntactic Pattern Recognition Using Finite Inductive Strings,” *The 5th IAPR International Conference on Pattern Recognition in Bioinformatics (IAPR PRIB 2010)*, Lecture Notes in Computer Science (LNCS) 5780, pp. 125–136, September 2010.
- [40] **Jinsuk Baek**, Paul S. Fisher, and Minho Jo “An Enhancement of mSCTP Handover with an Adaptive Primary Path Switching Scheme,” *Proceedings of IEEE Vehicular Technology Conference (IEEE VTC)*, September 2010.
- [41] Paul S. Fisher, John O. Adeyeye, **Jinsuk Baek**, and Joao Carlos Setubal, Finite Inductive Sequences, Kolmogorov Complexity with Application to Genome Sequences”, *Proceedings of International Conference on Bioinformatics, Computational Biology, Genomics and Chemoinformatics (BCBGC-10)*, pp. 78–83, July 2010.
- [42] **Jinsuk Baek**, Cheonshik Kim, Paul S. Fisher, Hongyang Chao, “(N, 1) Steganography Approach for Secret Sharing with digital images”, *Proceedings of IEEE International Conference on Wireless Communications, Networking and Information Security (IEEE WCNIS2010)*, pp. 325–329, June 2010.
- [43] Yong woon Ahn, Albert M. K. Cheng, **Jinsuk Baek**, and Paul Fisher, “Detection and Punishment of Malicious Wireless Stations in IEEE 802.11e EDCA Network,” *Proceedings of the 2010 IEEE Sarnoff Symposium (IEEE Sarnoff 2010)*, pp. 1–5, April 2010.
- [44] Paul S. Fisher, **Jinsuk Baek**, Minho Jo, “Program Readability: A Proposed Software Metric,” *Proceedings of the 2009 International Conference on Internet (ICONI & APIC-IST 2009)*, December 2009.
- [45] Paul S. Fisher, Howard Fisher, and **Jinsuk Baek**, and Cleopas Angaye, “Syntactic Pattern Recognition Using Finite Inductive Strings,” *Proceedings of the 4th IAPR International Conference on Pattern Recognition in Bioinformatics (IAPR PRIB 2009)*, Lecture Notes in Computer Science (LNCS) 5780, pp. 89–101, September 2009.
- [46] **Jinsuk Baek**, Min Gyung Kwak, Paul S. Fisher, and Elva J. Jones, “A Web Object Management Policy for Cooperative Hybrid Caching Architecture,” *Proceedings of the 2nd IEEE International Conference on Computer Science and Information Technology (IEEE ICCSIT 2009)*, pp. 61–65, August 2009.
- [47] **Jinsuk Baek**, Sun Kyong An, Paul S. Fisher, and Elva J. Jones, “Dynamic Cluster Header Selection with Self-incentive for Wireless Sensor Networks,” *Proceedings of the 2009 IEEE Sarnoff Symposium (IEEE Sarnoff 2009)*, pp. 1–5, March-April 2009.
- [48] **Jinsuk Baek**, Paul S. Fisher, and Mingyung Kwak, “FI-based File Access Predictor,” *Proceedings of the 49th ACM Southeast Conference (ACM SE 2009)*, pp. 1–4, March 2009.
- [49] **Jinsuk Baek**, Derrick Hargrove, Paul S. Fisher, and Cheonshik Kim, “An Urgent NAK with Prediction of Lost Packets for Fast Error Recovery in a Reliable Multicast,” *Proceedings of the 2008 International Workshop on Ubiquitous Convergence Technology (IWUCT 2008)*, pp. 113–120, August 2008.
- [50] **Jinsuk Baek**, Paul S. Fisher, Mingyung Kwak, and Hyungoo Han, “Secure Scalable and Reliable Multicast with FI-Based Local Group Key,” *Proceedings of the 2008 International Conference on Internet Computing (ICOMP 2008)*, pp. 188–194, July 2008.
- [51] **Jinsuk Baek**, Paul S. Fisher, Yuri Kim, Mingyung Kwak, and Hyungoo Han, “An Object Discarding Policy with FI Systems for Cooperative Hybrid Caching Architecture,” *Proceedings of the 46th ACM Southeast Conference (ACM SE 2008)*, pp. 411–416, March 2008.
- [52] YouSik Hong, Cheonshik Kim, and **Jinsuk Baek**, “A Study on the Traffic Accident Examination Using Artificial Intelligence,” *Proceedings of the 2007 International Ubiquitous*

- Conference/Workshop for Wireless and Ubiquitous Systems (IUC/IUW 2007)*, pp. 41–46, July 2007.
- [53] **Jinsuk Baek** and Wood Kanampiu, “An Admission Control Scheme for Repair Server in a Tree-Based Reliable Multicast,” *Proceedings of the 45th ACM Southeast Conference (ACM SE 2007)*, pp. 191–196, March 2007.
 - [54] Eunjung Lee, **Jinsuk Baek**, and Stephen S.-H Huang, “A Dynamic Mobility Management Scheme for VoIP Services,” *Proceedings of the 3rd International Conference on Information Technology: Next Generation (ITNG 2006)*, pp. 340–345, April 2006.
 - [55] **Jinsuk Baek** and Jehan-François Pâris, “A Tree-Based Reliable Multicast Scheme Exploiting the Temporal Locality of Transmission Errors,” *Proceedings of the 24th IEEE International Performance Computing and Communications Conference (IEEE IPCCC 2005)*, USA, pp. 275–282, April 2005.
 - [56] **Jinsuk Baek** and Jehan-François Pâris, “An Efficient Retransmission Control Scheme for Tree-Based Reliable Multicast,” *Proceedings of the 2004 International Symposium on Performance Evaluation of Computer and Telecommunication Systems (SPECTS 2004)*, USA, pp. 145–152, July 2004.
 - [57] **Jinsuk Baek** and Jehan-François Pâris, “A Heuristic Buffer Management Scheme for Tree-Based Reliable Multicast,” *Proceedings of the 4th IEEE International Symposium on Computer Communications (IEEE ISCC 2004)*, pp. 1123–1128, June-July 2004.
 - [58] **Jinsuk Baek** and Jehan-François Pâris, “An Efficient Buffer Management Scheme for Tree-Based Reliable Multicast Using Infrequent Acknowledgements,” *Proceedings of the 23rd IEEE International Performance Computing and Communications Conference (IEEE IPCCC 2004)*, pp. 13–20, April 2004.
 - [59] **Jinsuk Baek** and Eunjung Lee, “An Improved Logical Tree Construction Scheme for Reliable Multicast,” *Proceedings of the 11th IFIP International Conference on Telecommunication Systems (IFIP ICTS 2003)*, pp.110–121, October 2003.
 - [60] **Jinsuk Baek**, “A New Configuration Scheme for ACK tree in Reliable Multicast Protocols,” *Proceedings of the 2002 International Conference on Computer Communications (ICCC 2002)*, pp. 972–987, August 2002.
 - [61] **Jinsuk Baek**, “A Hybrid Configuration of ACK tree for Multicast Protocol,” *Proceedings of the 2002 International Symposium on Performance Evaluation of Computer and Telecommunication Systems (SPECTS 2002)*, pp. 852–856, July 2002.
 - [62] **Jinsuk Baek** and Sangchul Kim, “An Efficient MF (Model-Failure) Object Coding Method for Object-Oriented Coder,” *Proceedings of the 24th KISS Fall Conference*, pp.131–134, November 1997.
 - [63] Heekyung Kim, **Jinsuk Baek**, and Sangchul Kim, “CAI System for C and UNIX Using a Hypermedia Technology,” *Proceedings of the 22nd KISS Spring Conference*, pp.1063–1066, April 1996.

Dissertation, Thesis and Miscellaneous

- [1] **Jinsuk Baek**, “Scalable Approaches for Tree-Based Reliable Multicast”, Ph.D. dissertation, University of Houston, December 2004.
- [2] **Jinsuk Baek**, “An Efficient Compression Method for Broadcasting Protocol in Low Bit Transmission Rate,” *Bronze Prize for Good Research Paper, 1st SAMSUNG Electronics New Technology Research Contest*, December 2000.

- [3] **Jinsuk Baek**, “An Efficient Parameter Coding Method for Object-Oriented Coder (MPEG-4)”, Master thesis, Hankuk University of Foreign Studies, February **1998**.

Grant Proposals

- [1] “Winston-Salem Advanced Quantum Training, Education & Readiness (WAQTER) Program, Submitted to U.S. Army Research Office, LPS Qubit Collaboratory (LQC), (Co-PI, Submission Date: April 30, 2026, Amount: \$2,000,000).
- [2] “AI-Enabled Smart Manufacturing Platform for Real-Time 3D Electrode Microstructure Analysis in Lithium Metal Battery Manufacturing,” Submitted to U.S. Department of Energy (DoE) Genesis Mission: Transforming Science and Energy with AI, Phase I, (PI, Submission Date: April 28, 2026, Amount: \$388,583, WSSU portion: \$100,493).
- [3] “A Novel Low-Cost Robotic Assistant Integrating AI, 3D Printing, Embedded Microcomputer Programming, Sensor-Based Control, and Security for STEM Education,” 2025 WSSU Catalyst Program for Research Funding, Submission Date: October 13, 2025, (Co-PI, **Funded Amount: \$7,500**).
- [4] “Acquisition of an Advanced Collaborative Robotic Arm for Interdisciplinary Research and Teaching at WSSU,” 2025 WSSU Equipment Grant Program, Submission Date: September 9, 2025, (PI, Amount: \$13,868, Not Funded).
- [5] “Southeast Quantum Innovation Engine (SEQIE),” National Research Foundation (NSF), (PI, Submission Date: April 15, 2025, Amount: \$1,280,186, Not Funded).
- [6] “Cooking Robot System for Everybody,” Korea Ministry of Agriculture, Food and Rural Affairs, (PI, 2024-2027, **Funded Amount: \$3,000,000, WSSU Portion: \$137,000**).
- [7] “Innovate Education and Research Training for Quantum Computing (QC): Empowering Underrepresented Minority Students (UMS),” Submitted to U.S. Department of Education (DoEd), (PI, Submission Date: May 28, 2023, Amount: \$899,997, Not Funded).
- [8] “Expanding Quantum Research and Education at Winston-Salem State University with Research on Hybrid Microwave-Optical Quantum Devices,” National Research Foundation (NSF), (Senior Personnel, 2023-2028, **Funded Amount: \$5,000,000**).
- [9] “Training a Diverse Quantum Workforce in Carolinas through Establishing the Winston-Salem Quantum Education Collaboratory (WS-QEC),” Department of Energy (DoE) (Co-PI, 2023-2026, **Funded Amount: \$2,253,963**).
- [10] “Adjusting Level of Simultaneous Updates Using a Set Configuration CoAP Scheme for Internet of Things,” Submitted to IBM Center of The Business of Government for Research Stipend Program, (PI, Submission Date: March 25, 2020, Amount: \$20,000, Not Funded).
- [11] “Middleware API Development for Heterogeneous Device Management for IoT,” Submitted to National Research Foundation (NSF), (Co-PI as an invited scholar, Submission Date: November 2017, Not Funded).
- [12] “Asynchronous, Parallel, Plastic, Multi-Processing System (APPMPS): The Memory System,” Submitted to Department of Defense (DoD), (Co-PI, Submission Date: January 2015, Amount: \$593,661.00, Not Funded).
- [13] “Development of an Extensible File Synchronization System across Multiple, Heterogeneous Computing Devices,” Submitted to Department of Defense (DoD), (PI, Submission Date: January 2015, Amount: \$266,070.00, Not Funded).
- [14] “A Lightweight Transport Protocol to Support Reliable N-Screen Services,” Research Initiation Program (RIP) 2013 sponsored by WSSU, (PI, funded with **\$10,000**).
- [15] “Mobile Wireless Computer and Communication Systems and IT-BT Convergence for Education,” Submitted to Department of Defense (DoD), (PI, Submission Date: September 2010, Amount: \$90,658.00, Not Funded)

- [16] “Decision Support Model Based on Information Acquired by Dynamically Optimized Sensors-to-Commander Communications,” Submitted to Department of Defense (DoD), (Co-PI, Submission Date: October 2010, Amount: \$1,993,300, **Recommended but not Funded**)
- [17] “Design and Implementation of (N, 1) Secret Sharing Approach Based on Steganography with Gray Digital Images,” Submitted to IBM Center for The Business of Government, (PI, Submission Date: August 2010, Amount: \$20,000.00, Not Funded)
- [18] “Operations of BS with Mobile Robot for Wireless Sensor Networks,” 2010-2011 NC Space Graduate Fellowship, (Research advisor, funded with **\$6,000**).
- [19] “Parallel Computing Infrastructure to Advance Research and Education Capabilities”, Submitted to National Research Foundation (NSF), (Co-PI, Submission Date: August 2009, Amount: \$306,272.00 for WSSU and total is around \$600,000.00 in conjunction with UNCG, Not Funded)
- [20] “A Wireless Protocol for Prediction of Next Service for an All-Mobile Environment,” Submitted to Army Research Office (ARO), (Co-PI, Submission Date: March 2008, Amount: \$368,215.00, Not Funded).
- [21] “Ad Hoc Network, Security and a Mobile Protocol,” Submitted to National Research Foundation (NSF), (PI, Submission Date: March 2008, Amount: \$448,386.00, Not Funded).
- [22] “Smart, Mobile, Ad Hoc, Sensor Networks for Utilization by the Modern War-Fighter Applied to Surveillance in Battlefield Space,” Submitted to the Army Research Office (ARO), (Co-PI, Submission Date: January 2008, Amount: \$799,904.00, **Recommended but not funded**).
- [23] “WSSU DURIP: WiFi, IEDs Video Systems and Education,” Submitted to Department of Defense (DoD), (Co-PI, Submission Date: August 18, 2007, Amount: \$127,136.23, Not Funded).
- [24] “A Dynamic Admission Control Scheme for Reliable Multicast,” 2007-2008 NC Space Graduate Fellowship, (Research advisor, funded with **\$5,000**).
- [25] “Performance Evaluation of Novel Hybrid Cache Architecture for Cooperative Web Caching,” Research Initiation Program (RIP) 2007 sponsored by WSSU, (PI, funded with **\$10,000**).
- [26] “A NAK Suppression Scheme for Group Communications Considering the Spatial Locality of Packet Losses,” 2006-2007 NC Space Graduate Fellowship, (Research advisor, funded **\$5,000**).
- [27] “Seamless Multicast for Mobile Hosts in Mobile IPv6 Networks,” Submitted to the Research Initiation Program (RIP) 2006 sponsored by WSSU, (PI, Submission Date: February 2006, Not Funded).
- [28] “Tablet PCs in Teaching and Learning,” Submitted to the 2006 HP Equipment Support Program. (Co-PI, Submission Date: February 2006, Not Funded).

Professional Activities

Member: IEEE, KSII, and IEEK

Director: IEEK Computer Society

Journal Editor:

Guest Editor: Journal of Real-Time Image Processing: Special Issue on Real-Time Data Hiding and Visual Cryptography (2018)

Guest Editor: International Journal of Distributed Sensor Networks: Special Issue on Data mining Techniques based on Wireless Sensor Networks (2017)
Security and Communication Networks (2017 – Present)

Guest Editor: International Journal of Security and Communication Networks: Special Issue on Emerging Security and Privacy Influencing Enterprise BYOD and Future Mobile Network (2017)

Guest Editor: Springer Journal of Real Time Processing: Special Issue on Real-Time Data Hiding and

Visual Cryptography (2015)
IEIE Transactions on Smart Processing and Computing (IEEK SPC) (2012 – Present)
International Journal of Computer Science and Network Security (IJCSNS) (2006 – Present)
The Smart Computing Review (Smart CR) (2011 – 2024)
KSII Transactions on Internet and Information Systems (TIIS) (2007 – 2024)

Conference Technical Program Committee:

APIC-IST 2025, SICBS 2017, BigComp 2015, AICT 2015, 3PGCIC 2015, IEEE IPCCC 2014, UMAS 2012, UMAS 2013, IEEE ICCCN 2011, IEEE VTC Fall 2010, ICONI 2009, IEEE ITNG 2007, ACM SE 2007, ACM SE 2008, IUC/IUW 2007, IUC/IUW 2008, ICUCT 2007

Journal Reviewer:

Discover Internet of Things Journal (2026 – Present)
Agriculture (2022 – Present)
Electronics (2022 – Present)
Sensors (2021 – Present)
IEEE Access (2020 – Present)
Springer Mobile Networks and Applications (2017 – Present)
Springer Personal Ubiquitous Computing (2017 – Present)
Hindawi Security and Communication Networks (2017 – Present)
Springer Journal of Real Time Processing (2015 – Present)
IEEE Wireless Communications (2014 – Present)
Journal of Applied Mathematics (2013 – Present)
Multimedia Tools and Applications (2013 – Present)
Hindawi International Journal of Distributed Sensor Networks (2013 – Present)
EURASIP Journal of Wireless Communications and Networking (2013 – Present)
IEEE Transactions on Circuits and Systems for Video Technology (2012 – Present)
IEEE Systems Journal (2012 – Present)
Springer Cluster Computing (2012 – Present)
Inderscience International Journal of Intelligent Information and Database Systems (2011 – Present)
IEEE Transactions on Broadcasting (2010 – Present)
Elsevier Computer Networks (2010 – Present)
Wiley International Journal of Communication Systems (2010 – Present)
IEEE Transactions on Mobile Computing (2009 – Present)
Hindawi Journal of Computer Systems, Network, and Communications (2009 – Present)
Wiley Wireless Communications and Mobile Computing (2008 – Present)
Journal of Supercomputing (2008 – Present)
IEEK Journal (2008 ~ Present)
KSII Transactions on Internet and Information Systems (TIIS) (2007 – Present)
Journal of Ubiquitous Convergence Technology (2007 – Present)

Conference Reviewer:

SICBS 2017, DemSet 2012-2014, ISA 2012, ACM ICUIMC 2011, CCCT 2011, IEEE CHINACOM 2009, IEEE GLOBECOM 2004, IEEE IPCCC 2004 and 2005.

Services

Department Level (Department of Computer Science at WSSU)

- CS Faculty Classroom Observation Chair (Fall 2025, Spring 2026)
- CS Faculty Search Committee Chair (Spring 2025)
- HBCU Smart Cities Challenge Advisor for Team WSSU (Fall 2022)
- CS Faculty Representative for WSSU Open House (Fall 2022 – Spring 2026)
- CS Post-Tenure Review Committee Chair (Spring 2017, Fall 2022)
- CS Faculty Reappointment Committee Chair (Spring 2022)
- CS Faculty Meeting Recorder (Fall 2007 – Spring 2008, Fall 2020 – Spring 2021)
- CS Faculty Search Committee (Spring 2010, Spring 2023 – Spring 2024)
- Recruitment & Retention Committee (Fall 2008 – Spring 2010)
- Department Enrichment Committee (Fall 2007 – Present)
- Graduate Admission Committee (Fall 2006 – Present)
- Undergraduate Program Curriculum/Assessment Committee (Fall 2006 – Spring 2010)
- Graduate Program Curriculum/Assessment Committee (Fall 2006 – Present)
- International Initiative for Foreign Students, South Korea (Fall 2005 – Present)

University Level (Winston-Salem State University)

- WSSU Professional Development Committee (Fall 2017 – Present)
- Have recruited 17 International Students from South Korea (Fall 2006 – Present)
- WSSU Freshman General Education Advisor (Fall 2012 – Present)
- WSSU IAPP India Planning Task Force (Spring 2010 – Present)
- WSSU Globalization Task Force (Fall 2009 – Present)
- WSSU International Program Committee (Fall 2009 – Present)
- WSSU Faculty Senate (Fall 2006 – Present)
- WSSU Academic Standards and Curriculum Committee (Fall 2008 – Present)
- WSSU Constitution Committee (Fall 2008 – Spring 2010)
- WSSU Graduate Council (Fall 2007 – Spring 2009)
- Search Committee for International Programs Advisor (Fall 2008)